

PROJECT (TEAM WORK)

**ELECTRONIC AND OPTOELECTRONIC
INSTRUMENTATION**

MASTER IN TELECOMMUNICATION ENGINEERING

1st Course, 2nd Semester

Academic Year 2025-2026

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DEPARTAMENT OF ELECTRONICS TECHNOLOGY

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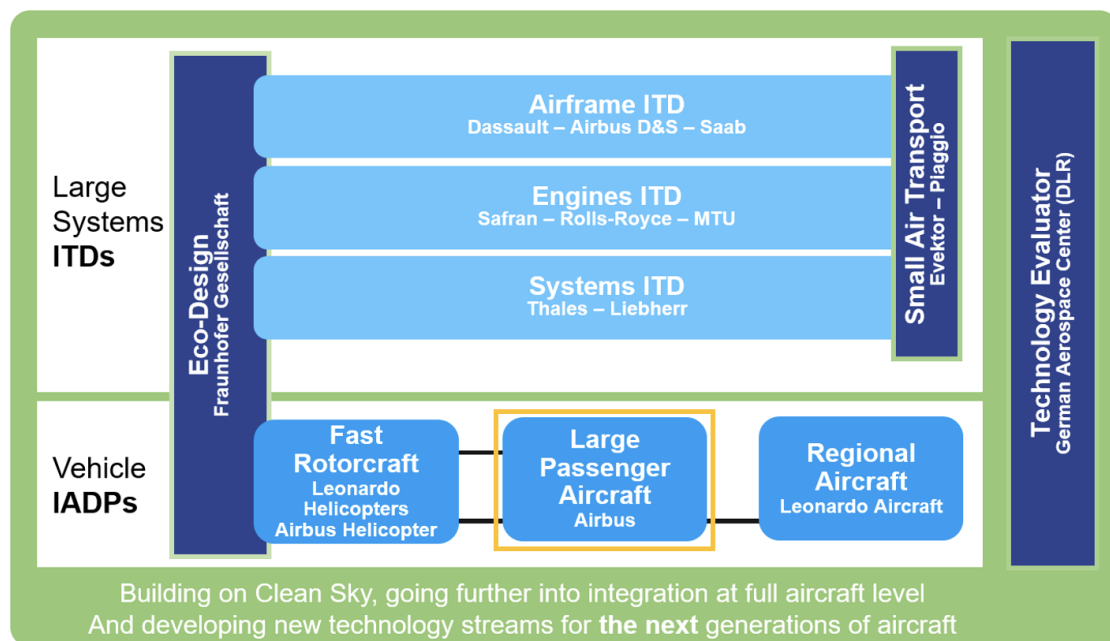
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INSTRUMENTATION WITH MEMS MICROPHONE ARRAYS FOR AERONAUTICAL TESTING (CLEANSKY 2, H2020, EU)

1. MOTIVATION AND OBJECTIVES

This challenge is part of the H2020 framework program of the European Union, within CleanSky 2, LPA (Large Passenger Aircraft) program area. The main contractor is AIRBUS and the technology evaluator is DLR. The call H2020-CS2-CFP11-2020-01 proposes the topic [JTI-CS2-2020-CFP11-LPA-01-88](#): *Development of New digital Microphone-MEMS-Sensors for wind tunnels with open/closed test sections and flight tests.*



Type of action (RIA/IA/CSA)	IA	Horizon 2020 Clean Sky 2 – Large Passenger Aircraft IAPD	
Program Area	LPA	 	
(CS2 JTP 2015) WP Ref.	WP 1.1		
Indicative Funding Topic Value	1400 k€		
Topic Leader	Airbus		
Type of Agreement	Implementation Agreement		
Duration of the action	30 Months		
Indicative Start Date	> Q4 2020		
Topic Identification Code	JTI-CS2-2020-CfP11-LPA-01-88		
Title	Development of New digital Microphone-MEMS-Sensors for wind tunnels with open/closed test sections and flight tests		
Short description	The main objective is to develop a novel technology for surface unsteady pressure measurements which are suitable for arbitrary model/aircraft position (including cockpit area) and with high spatial resolution. State of the art microphones are too large to capture the unsteady pressure fluctuations underneath the turbulent boundary layer to predict the cabin noise excitation (especially in regions with high pressure gradients). MEMS sensors seem to be a good candidate to overcome. State of the art microphones have limitations. This topic aims at developing new digital MEMS based microphones fulfilling acoustic requirements for wind-tunnels and flight test applications.		

The objective of our participation is to carry out the design of the electronic instrumentation system that allows the use of MEMS microphone arrays in aeronautical tests. The requirements that must be met are summarized in the following table included in the call (Table 1):

Table 1: General requirements for aero-acoustic MEMS microphones in FT and WTT

Criterion		Wind tunnel test application <i>Open test section</i>	Wind tunnel test application <i>Closed test section</i>	Fligh test application
Airframe noise application Dynamic Range [linear dB ref. $2 \cdot 10^{-5}$ Pa] (integrated rms value)	Min	0	30	45
	Max	140	150	145
Rotor/Fan noise application Dynamic Range [linear dB ref. $2 \cdot 10^{-5}$ Pa] (integrated rms value)	Min	30	30	45
	Max	170	170	175
Frequency Response [Hz] (see footnote 8 and remarks)	+/- 3 dB	100 – 35000	100 – 100000	40 - 10000
	-12dB/oct. ⁸	500 – 50000	n/a	n/a
Signal to Noise ratio below 120 dB [dB]		90	90	70
Temperature range [°C]	Min	-50	-20	-65
	Max	70	70	30
Ambient pressure stability range [Pa] ⁹	Min	99000	99000	22000
	Max	102000	102000 (not pressurized) 500000 (pressurized)	102000

We will address the requirements of two applications at the same time on the base of a new sensing technology. The applications are flight tests (FT) and wind tunnels tests (WTT) in a closed section, corresponding to the last and penultimate columns, respectively.

The requirements to be met are the following: measurement range, frequency response, signal-to-noise ratio, and operating temperature range, corresponding to the first five rows of the table. As an additional requirement, we will power all conditioning electronics from the aircraft DC bus (28 V) or from an auxiliary battery (24 V).

2. AVAILABLE MEANS

We have a new technology (dual-frequency) of piezoelectric (AlScN) MEMS sensors developed for the Aeromic project. Its characteristics are presented in Table 2.

Table 2: Characteristics of Aeromic piezoelectric MEMS sensors

Metric	Target value
Dynamic range	140 dB (30-170 dB)
Bandwidth	100 Hz – 100 kHz
Sensitivity	5-10 mV/Pa

These dual-frequency sensors are an evolution from four identical isosceles triangular cantilevers, as developed for general-purpose piezoelectric MEMS microphones, to a design with asymmetrical scalene triangular cantilevers (Fig. 1): two large ones covering the response in the low frequency range (up to 10 kHz, with 30 kHz resonance) and two small ones performing better at higher frequencies (up to 100 kHz, with 200 kHz resonance).

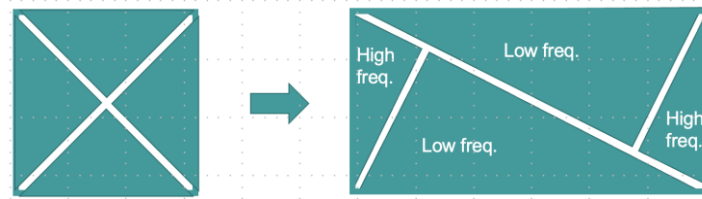


Figure 1: Evolution of the structure of cantilevers of the dual-frequency MEMS sensors

As a result, the MEMS microphones provide with two AC-coupled outputs of different frequency range (Fig. 2): lower frequency (LF) and higher frequency (HF). The sensitivity of the HF output is 5 mV/Pa, and the sensitivity of the LF output is 10 mV/Pa.

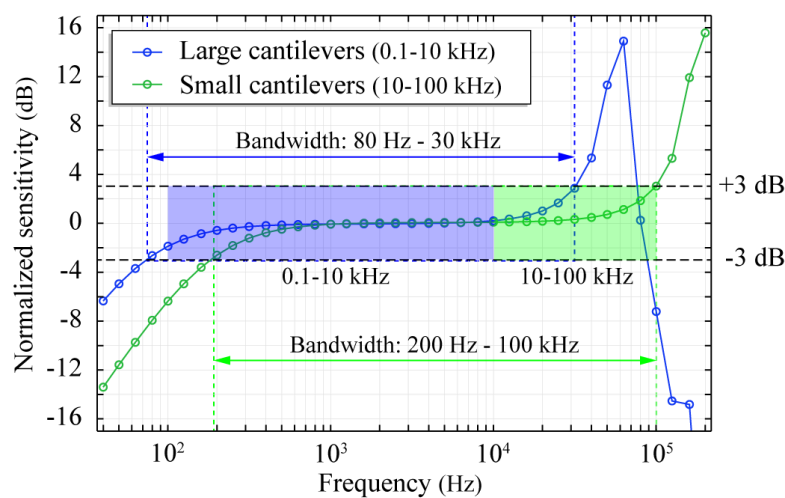


Figure 2: Frequency response of the dual-frequency MEMS sensors

The impedance of each output is complex (Fig. 3). However, we will assume the following values of the piezoelectric sensor equivalent circuit: capacitance 1.9 nF and resistance 90 k Ω for the LF section, and capacitance 785 pF and resistance 47 k Ω for the HF section.

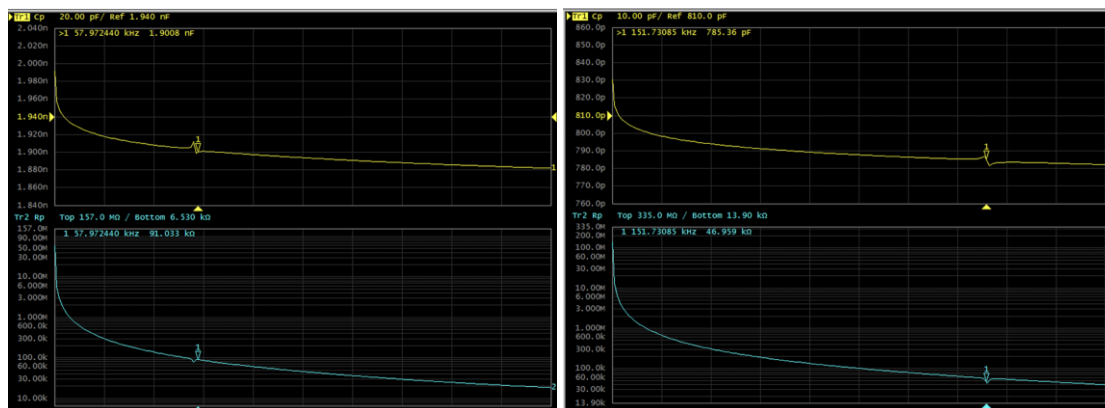


Figure 3: Impedance analysis of the dual-frequency MEMS sensors

The MEMS sensors are of reduced size (900 μm x 1450 μm), and can be manufactured and integrated in the form of a sensor array. The target is a microphone array of 80 MEMS sensors integrated in a flexible printed circuit board.

The electronic instrumentation system proposed by our competitors is divided into three blocks: Firstly, the analog signal conditioning electronics for 80 MEMS sensors; secondly, the concentration of the signal channels in a limited number of cables (shielded twisted pairs); and, finally, a remote DAQ system with analog inputs. Hardware time multiplexing and software demultiplexing are proposed. The design of the signal conditioning electronics uses commercial integrated circuits. Alternatively, a design can be chosen that provides a serial digital output for each MEMS on the flex PCB and a digital remote DAQ.

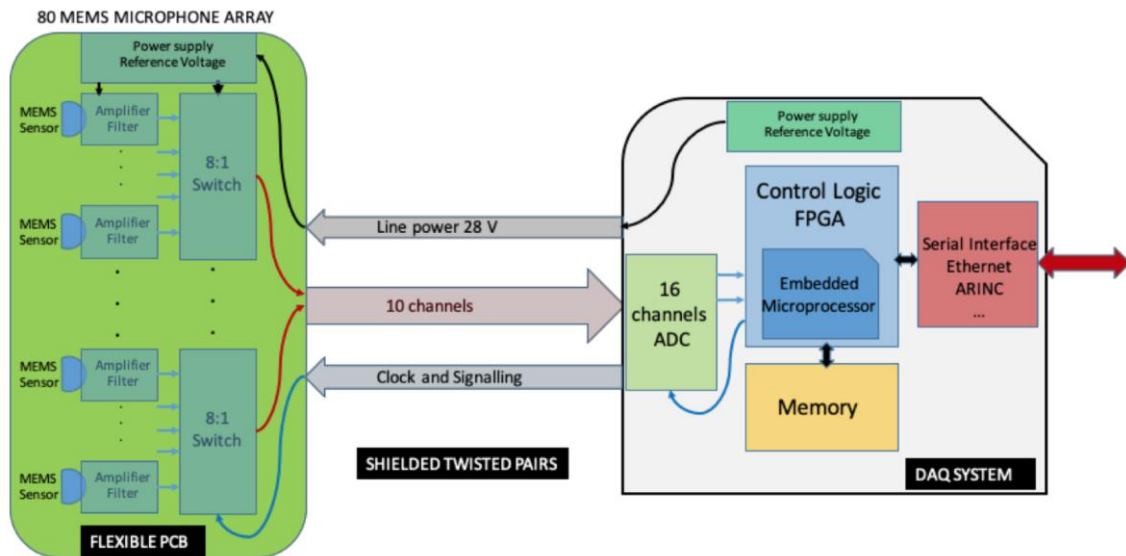


Figure 3: Block diagram of a proposal of conditioning system connected to the DAQ

3. **METHODOLOGY**

It is recommended to follow the methodology described below:

- 1) Block diagram of the complete system, describing the functionality of each block and specifying the interfaces between blocks.
- 2) Functional design of the signal conditioning of a MEMS sensor.
We can take as reference the case study provided.
- 3) Choosing the right components taking into account:
 - a. Technical characteristics vs. specifications
 - b. Operating conditions and their compatibility with the application
 - c. Dimensions for its integration in the flexible PCB (target 2 mm thick)
- 4) Design scaled to an array of 80 sensors.
 - a. Possible multiplexing
 - b. Wiring and connectors
- 5) Acquisition Strategy
Select a DAQ suitable for the application. We can take as a reference systems not designed to be on board an aircraft. Specify its characteristics (number of bits, acquisition frequency, number of channels, digital inputs, timers, etc.).
- 6) Systems integration/control
Select the software design to integrate and control the instrumentation system from a single user interface. Describe it at a high level in MATLAB, LABVIEW or PHYTHON, etc.

4. **FOLLOW-UP, DELIVERABLES AND SELF-ASSESSMENT**

To track your work, we recommend using a shared **portfolio** or repository.

At the end of the work, the delivery will consist of a joint **report** made by the entire team, containing the recommended sections in the methodology and the necessary annexes.

The report will include a **joint self-assessment** on the degree of compliance with specifications, the performance of the sensor system, and the proposed improvements. Furthermore, an **assessment of each member** of the team, indicating the role(s) performed, the individual contributions to the work, and the participation in joint sections.

Finally, an oral **presentation** will be made with the participation of all the members of the team, from which the support used will be delivered (PPT or similar, video, link to YouTube, etc.). The presentation can be recorded on video to be used in the face-to-face session.

We have the [evaluation matrix](#) as a reference to check the quality and completeness of the deliveries. See also the [document on collaborative teamwork](#).